HDOC- 6 – 1
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Program to Check Whether a Number is Even or Odd Using if–else

This document provides a clean, well-commented implementation for checking whether an integer is even or odd in C. The program takes an integer as input and uses the modulus operator with an if–else statement to decide and display the result.

1. C Implementation

Step 1: Understand the Logic

- A number is even if it is exactly divisible by 2, i.e. the remainder is 0.
- A number is odd if dividing it by 2 leaves a non-zero remainder.
- The modulus operator (%) is used to find the remainder.

Step 2: Define Variables and Data Types

- Use an integer variable (int) to store the input number.

Step 3: Take User Input

- Display a prompt using printf("Enter an integer: "), then read the value using scanf("%d", &num).

Step 4: Apply the if–else Condition

1. If num % 2 == 0, the number is even.
2. Otherwise (else), the number is odd.

Step 5: Format the Output

- Print the number along with the message "even number" or "odd number" using printf().

Complete C Code

```c
/*
 * Q6 (User Inputs, Operations & Output)
 * Write a program to input an integer and check whether
 * it is even or odd using if-else.
 */
#include <stdio.h>

int main() {
    int num;

    // Step 1: Read the integer
    printf("Enter an integer: ");
    scanf("%d", &num);

    // Step 2: Check the remainder when divided by 2
    if (num % 2 == 0) {
        printf("%d is an even number.\n", num);
    } else {
        printf("%d is an odd number.\n", num);
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc6.c -o hdoc6
$ ./hdoc6
Enter an integer: 14
14 is an even number.
```

Run 2:

```
$ ./hdoc6
Enter an integer: 7
7 is an odd number.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 14
- Calculation:
  - num % 2 = 14 % 2 = 0
  - Condition (num % 2 == 0) is TRUE, so the if block runs
- Output: 14 is an even number.

Sample Test Case 2:

- Input: 7
- Calculation:
  - num % 2 = 7 % 2 = 1
  - Condition (num % 2 == 0) is FALSE, so the else block runs
- Output: 7 is an odd number.